Estimating Bridge Clearance from Street-Level Imagery

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The Problem

HERE maps attributes like bridge vertical clearance, which matter for truck routing

Today they come from LiDAR surveys and field audits: expensive to repeat at network scale

Question: can existing crowd-sourced street-level imagery (Mapillary) estimate clearance in metres?

Ground truth: FHWA National Bridge Inventory, Item 54B (minimum vertical under-clearance)

Our Pipeline

NBI bridge record → Mapillary image sequence driving toward the bridge

Segmentation (Mask2Former): bridge underside + road surface

Metric 3D reconstruction with a swappable backbone: MapAnything / Depth Anything 3 / VGGT

Measure: RANSAC road plane → median distance to the bridge underside

Quality control + confidence decision (accept / review / reject) → GeoJSON output

Every backbone uses the same 80-bridge Chicago benchmark, frames, masks and scoring code

Who Did What

Shared: 80-bridge Chicago benchmark, frames, masks and scoring code

Kely: MapAnything backbone, error analysis, Pipeline v2, confidence rules, generalization tests

Anna: Depth Anything 3 backbone, Mask2Former vs SAM3 segmentation comparison

Adaora: VGGT backbone (GPS-based scale), building height (in progress)

MapAnything Results (Kely)

Single-view: median error 0.43 m, MAE 0.92 m on 30 quality-checked bridges; 50% within 0.5 m

Multi-view: MAE 1.09 m, less biased; paired difference vs single-view not significant

Phone / action-camera imagery: MAE 0.65 m

Metric scale is right: recovered camera height ≈ 1.3 m (car / handlebar mount)

Fast: about 6 seconds per bridge on one L4 GPU

Where the Errors Come From (Kely)

Camera: wide-angle dashcams (no EXIF) MAE 2.03 m vs 0.65 m for phones; lens distortion squashes the bridge

Distance: camera closer than 30 m → under-estimates by 1.35 m on average

Wrong target: a barrier or sign measured instead of the deck (estimates below 3 m)

All three can be detected from metadata or the estimate itself, without ground truth

Fixing Coverage: Pipeline v2 (Kely)

Biggest loss: 41 of 80 bridges had no bridge visible in the selected images

Cause: the NBI coordinate marks the structure, not the road that passes under it

Fix 1 (detection-guided retrieval): pick the image sequence by what the segmenter actually sees

Fix 2 (lens undistortion): correct each frame with Mapillary camera parameters

Coverage 40% → 62.5% (32 → 50 bridges) without changing the 3D model

Confidence Rules & Generalization (Kely)

Rules frozen on the benchmark: QC check, plausibility gate (3–9 m), camera rule, view agreement → accept / review / reject

51 held-out Chicago bridges: MAE 0.87 → 0.73 m after rules, 84% of estimates kept

120 bridges in Indianapolis & Milwaukee: accepted single-view MAE 0.70 m, 80% within 1 m (Chicago: 0.68 m)

Multi-view transfers worse (+1.1 m bias in Milwaukee) → single-view is the default

Honest Baseline & Value for HERE (Kely)

Always predicting the regional median (4.97 m) still has lower absolute error (0.42 vs 0.70 m)

But imagery ranks bridges: AUC 0.75 for low clearance (< 4.6 m), 0.90 for high (> 5.5 m)

Recommended use: screening + detecting wrong map values, not replacing surveyed values

The full Illinois inventory can be screened in a few GPU-hours

ANNA | Depth Anything 3 Results

[Outline for Anna to fill in]

How DA3 gets metres: metric depth + camera intrinsics → back-projection to 3D points

Coverage: single-view vs multi-view quality-checked bridges

Accuracy: MAE / median error on S1, S2, S3 (paper Table 1)

Visuals: example reconstruction (Figure 3) + predicted vs NBI scatter (Figure 11)

Takeaway: how DA3 compares with MapAnything

ANNA | Segmentation: Mask2Former vs SAM3

[Outline for Anna to fill in]

Why try SAM3: text prompts (“bridge underside”, “road”) instead of fixed classes

Trade-off: SAM3 had lower error but far fewer usable bridges

Paired comparison on identical bridges (Figure 12): difference not significant

Takeaway + next step: use SAM3 only as a fallback when Mask2Former finds nothing

Optional: spring DA3 + SAM3 experiments on IDOT sign structures